

# MARY LETEY

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## EDUCATION

|                       |                                |                                  |
|-----------------------|--------------------------------|----------------------------------|
| Sept 2023 — Present   | <b>Harvard University</b>      | PhD — <b>Applied Mathematics</b> |
| Sept 2022 — June 2023 | <b>Perimeter Institute</b>     | MSc — <b>Theoretical Physics</b> |
| Sept 2018 — June 2022 | <b>University of Cambridge</b> | BA Honours — <b>Mathematics</b>  |

## PUBLICATIONS & MANUSCRIPTS

\* denotes equal first authorship; † denotes equal senior authorship.

In-context classification: Calibration and noise in linear attention with softmax readout

*Under review* • Rysmakhanov\*, **M. Letey**\*, Lu, Zavatone-Veth, Pehlevan

One-step spectral dynamics of the neural tangent kernel under feature learning

*Under review* • Rysmakhanov, **M. Letey**, Pehlevan

What can linear attention learn in-context from nonlinear teachers?

*Under review* • **M. Letey**\*, Rysmakhanov\*, Lu, Pehlevan, Zavatone-Veth

Sequential correlations change in-context learning: Effective context length and architectural mismatch

[HiLD Workshop at ICML 2026](#) • **M. Letey**, Lu, Pehlevan†, Zavatone-Veth†

Pretrain–test task alignment governs generalization in in-context learning

[ICLR 2026](#) • **M. Letey**, Zavatone-Veth, Lu, Pehlevan

Theory of scaling laws for in-context regression: Depth, width, context and time

[ICLR 2026](#) • Bordelon, **M. Letey**, Pehlevan

Asymptotic theory of in-context learning by linear attention

[Proceedings of the National Academy of Sciences \(2025\)](#) • Lu, **M. Letey**, Zavatone-Veth, Maiti, Pehlevan

Quantum initial conditions for curved inflating universes

[Physical Review D \(2024\)](#) • **M. Letey**\*, Shumaylov\*, Agocs, Handley, Hobson, Lasenby

Discrete symmetry in Lorentzian spaces

[Master’s thesis, Perimeter Institute \(2023\)](#) • Advisor: Latham Boyle

## INVITED TALKS

**Johns Hopkins** — Data Science Seminar, Fall 2026 — Invited speaker

**EPFL** — [FLARE Workshop](#), May 2026 — Invited talk on theoretical models of in-context learning

**SIAM** — Conference on Uncertainty Quantification, March 2026 — Invited talk: “Asymptotic theory of in-context learning”

**Kempner Institute** — [Spring Into Science Retreat](#), May 2024 — Opening talk: “Asymptotic theory of in-context learning”

## WORK & RESEARCH EXPERIENCE

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|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Research Internship<br>Winter 2026             | <b>Hudson River Trading</b> • New York, USA<br>Machine learning for market prediction                                  |
| Research Internship<br>Summer 2023             | <b>Quebec Artificial Intelligence Institute</b> • Montreal, Canada<br>Equivariant architectures for neural PDE solvers |
| Research Internship<br>Summer 2022             | <b>Cavendish Laboratory</b> • Cambridge, UK<br>Research on early-universe cosmology and curved spacetimes              |
| Internship<br>Summer 2020                      | <b>Embryo Ventures</b> • London, UK<br>Due diligence and founder outreach                                              |
| Research Internship<br>Fall 2019 — Spring 2020 | <b>Iguana Investments</b> • Cambridge, UK<br>Market prediction models                                                  |

## AWARDS & FUNDING

|      |                                    |                                                       |
|------|------------------------------------|-------------------------------------------------------|
| 2025 | <b>Graduate Fellowship</b>         | Kempner Institute — Full support for graduate studies |
| 2023 | <b>Graduate Prize Fellowship</b>   | Harvard University                                    |
| 2022 | <b>International Scholar Award</b> | Perimeter Institute — Full support for MSc            |